Night Signals

A public synthetic-data audit and internal model evaluation

Source and scope

The primary file contains 1,000 public Kaggle records that its uploader describes as synthetic. This project did not design or conduct a survey. No sampling frame, original item wording, generation algorithm, or within-night timestamps are provided. Results describe this file, not a sampled population. Source: https://www.kaggle.com/datasets/harpartapsingh13/sleep-and-doomscrolling-habits-dataset. Local source checksum and field caveats appear in `data/PROVENANCE.md` and `DATA_DICTIONARY.md`.

Target validity

A category-by-score check finds that 213 of 330 records labeled Poor have `sleep_quality_score`=5/5. The label derivation is undocumented. The classifier recovers a synthetic-file label; it is not evidence of detecting a sleep disorder in people.

An arbitrary sensitivity label, score < 5, agrees with the Poor label in 75.5% of 960 complete records (Cohen's $\kappa = 0.355$). This exposes label dependence; neither label is clinically validated.

Exploratory association checks

Five checks were specified after earlier dataset inspection; they are not preregistered. Pearson chi-square uses complete cases; Benjamini–Hochberg FDR adjusts the five comparisons. Cramer's V is descriptive. Significance alone cannot establish a protective effect.

Variable	n	Cramer's V	FDR p
bedtime routine type	980	0.097	0.0464
uses night mode	1000	0.034	0.705
keeps phone in bedroom	1000	0.060	0.273
consumes negative news content	1000	0.024	0.742
doomscroller label	1000	0.457	2.35×10^{-45}

Routines and negative news

The unadjusted routine association is small. Reading does not stand out as protective: its Good share is below the 34% file-wide rate. The no-fixed-routine group has the highest Good share. Negative-news exposure has a weak association with sleep category ($p \approx 0.742$). These are descriptive contrasts.

An exploratory complete-case logistic adjustment ($n = 953$) included age, gender, occupation, bedtime exposure, doomscroller label, negative-news exposure, night mode, and phone location. No routine term survives seven-term FDR adjustment. These contrasts cannot identify protective habits.

Routine	n	Good (%)	Poor (%)
Meditation/Journaling	108	36.1	38.0
No Fixed Routine	156	42.9	24.4
Reading	157	31.8	33.1
Scrolling Social Media	375	28.0	38.1
Watching Videos	184	37.0	29.3

Model protocol

The binary target is Poor versus Good/Fair. Measured sleep outcomes, quality score, and record ID are excluded. Imputation, scaling, and categorical encoding occur within model pipelines. Candidate comparison uses five outer and three inner cross-validation folds on 750 development records. A sigmoid-calibrated logistic ensemble and cost-weighted threshold were fit from development data. Earlier experiments may have inspected overlapping holdout rows, and the cross-sectional source cannot prove predictor availability at bedtime.

Internal evaluation

Selected logistic model nested-CV balanced accuracy: 0.737 (fold SD 0.041). Historical 250-row holdout: balanced accuracy 0.760 (bootstrap 95% interval 0.704–0.816), ROC AUC 0.825, PR AUC 0.720, Brier 0.155, sensitivity 0.723, specificity 0.796; threshold 0.335. These are internal label-recovery metrics.

Retrospective holdout calibration: intercept 0.149 (95% CI −0.231 to 0.529); slope 1.218 (95% CI 0.889 to 1.546). This is not external calibration.

Development-only comparator	Features	BA at 0.5	OOF ROC AUC
majority class	0	0.500	0.500
doomscroller label rule	1	0.713	0.713
bedtime minutes only	1	0.651	0.755
full minus self-label and phone checks	20	0.676	0.746
full minus self-label	21	0.682	0.757
full candidate features	22	0.694	0.775

The fixed-threshold logistic comparisons are diagnostic and are not directly equivalent to the tuned and calibrated nested-CV production estimate.

On identical development rows at the same threshold, full untuned logistic minus the simple doomscroller rule has balanced-accuracy difference −0.019 (stratified paired bootstrap 95% CI −0.052 to +0.014). The full model's ROC AUC is 0.062 higher (95% CI +0.035 to +0.091), but the binary rule offers limited ranking resolution. Incremental classification benefit is unestablished.

Independent context and limitations

NHANES includes 8,040 analytic adults. MEC-weighted, age/sex-adjusted WLS with stratum-centered masked-PSU Taylor variance gives adjusted short-sleep systolic difference 0.82 mmHg (95% CI −0.86 to 2.50; $p = 0.316$) and diastolic difference 1.93 mmHg (95% CI 0.67 to 3.19; $p = 0.005$). Adult complete-case domain analysis remains approximate. PAD800/PAD820 are minutes per occasion; the activity comparison does not support a protective claim. PhysioNet subjects are unrelated to the Kaggle file. Neither dataset validates the classifier. High-exposure favorable-sleep cases are few; outcome contrasts are partly circular. External real-world model validation has not been performed.

Recompute the inputs with the scripts documented in `README.md`. This TeX file is the editable report source. Older figures remain historical.